

Christopher Tastad

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DATA SCIENCE LEADERSHIP SUMMARY

Senior data management leader with 8+ years architecting enterprise analytics platforms and establishing data governance frameworks across 0.5PB+ genomic datasets. Expert in FAIR-compliant data distribution, regulatory compliance (HIPAA, NIH), and AI/ML platform development. Proven track record of leading cross-functional teams, implementing automated analytics pipelines, and delivering strategic insights that drive \$9.5M+ in organizational value through data-driven decision making and operational optimization.

DATA SCIENCE EXPERIENCE

Lead Data Engineer

Feb 2021 – Present

NIDDK IBD Genetics Consortium

New York, NY

- Direct data governance across 650+ global collaborators spanning dozens of international consortia, ensuring HIPAA compliance and streamlined data sharing protocols for multi-site genomic studies at unprecedented scale.
- Lead consortium data team of 4 bioinformaticians and data scientists, directing technical strategy, coordinating collaborative analysis efforts, and establishing standardized methodological approaches for consortium-wide genomic studies and data integration initiatives.
- Implemented FAIR-compliant data distribution architecture using Open Science Framework, fundamentally improving accessibility and meeting NIH data sharing requirements for 0.5PB+ genomic datasets across international research community.
- Engineered comprehensive ID harmonization platform (IDhub) unifying legacy sample/subject IDs across 20 years of consortium recruitment, developing PostgreSQL database, REST API, global subject ID resolution service, and multi-staged curation workflows with audit logging.
- Deployed ultra-fast, web-based genomic annotation platform in AWS for WES/WGS analysis, reducing annotation time from days to hours for consortium-wide sequencing projects, making detailed genomic insights fast and widely accessible.
- Led comprehensive reorganization of 20 years of consortium study data across multiple eras of data collection, projects, and institutional partnerships, establishing systematic data stewardship at an international scale.
- Established cloud-hosted automated subject identification system serving as central source of truth for longitudinal study recruitment and data collection, creating systematic data governance where none existed.
- Implemented comprehensive data governance architecture focused on isolating PHI data within respective compute and storage environments, preventing contamination and ensuring compliance across institutional boundaries.
- Developed automated metadata management systems using LabKey sample management API, reducing manual curation overhead while establishing systematic data stewardship practices.
- Replaced \$250K+ annual vendor contracts with cost-effective open-source solutions, meeting and exceeding data processing and hosting requirements while demonstrating strategic resource optimization.

Senior Manager, Bioinformatics

Oct 2023 – Present

Icahn School of Medicine at Mount Sinai, Cho Lab

New York, NY

- Lead 5-person bioinformatics team driving technical strategy for cutting-edge genomic analysis across 10+ concurrent multi-year projects, establishing collaborative frameworks where none previously existed.
- Serve as tech lead on all methodological innovation, establishing leading analytical frameworks across dozen+ major project initiatives including evolving single-cell processing, automated cell typing, multi-omic data, spatial transcriptomics, and emerging genomic platforms.
- Pioneer spatial transcriptomics methodology extending Sopra for highly scalable execution and developing Parallax pipeline for noise correction, context-aware spatially differential expression, and downstream spatial analysis, establishing standard for foundational analytical frameworks.
- Implement comprehensive data lifecycle management including storage class transition policies, 3-2-1 backup paradigm, data version control (DVC), and project-specific data registries. Achieve zero data loss incidents over 4+ years while dramatically reducing technical debt across 0.5PB+ managed data.
- Selected as 10x Genomics beta partner for cloud hosting feature development in Xenium Explorer analysis application, directly influencing product roadmap through technical feedback and validation of cloud-specific hosting capabilities.
- Developed official Mount Sinai institutional execution profile for nf-core pipelines, creating standardized HPC configuration endorsed by core scientific computing team and accepted as open-source contribution to nf-core community codebase.
- Implemented enterprise Sourcegraph deployment and served as admin for secure team LLM access, pioneering integration of AI-assisted development tools in bioinformatics workflows and establishing institutional standards for responsible AI adoption in genomic analysis.

Bioinformatician II

Feb 2021 – Oct 2023

Icahn School of Medicine at Mount Sinai, Cho Lab

New York, NY

- Contributed to \$9.5M+ grant awards through technical leadership, experimental design, and strategic vision, providing data management expertise across all consortium projects and influencing methodological approaches at international scale.

- Established standardized single-cell preprocessing paradigm (CoreSC) creating unified statistical methodology across 15+ projects where none previously existed, fundamentally improving reproducibility and eliminating analysis methods inconsistencies.
- Constructed comprehensive scientific analysis application stack including containerized workflows, automated data distribution, and several multi-user self-hosted services, fundamentally lowering barriers to access and accelerating discovery outcomes.
- Pioneered internal containerization paradigm using Docker/Kubernetes for application deployment, enabling non-technical users to deploy reproducible analysis environments in high performance compute resources.

Graduate Research Assistant

May 2019 – Feb 2021

University of Minnesota Dept. of OBGYN, Starr Lab

Minneapolis, MN

- Applied advanced bioinformatic analysis to copy number variation on largest single-cell sequencing human cancer dataset of its kind, developing novel analytical approaches for genomic variation detection.
- Developed automated analysis pipeline from raw data to publication-ready figures, incorporating Hidden Markov Models (HMM) and FastCall algorithms for statistical genomic analysis.
- Executed full rewrite of cancer gene web application, deploying new ETL framework and reducing technical debt while improving system performance and data accessibility.

Graduate Research Assistant

Jan 2016 – Aug 2017

National Marrow Donor Program, Bioinformatics

Minneapolis, MN

- Served as data scientist in corporate R&D department focusing on hematological cancers, contributing to IP generation initiatives using 13 million patient database for predictive analytics and clinical insights.
- Developed integrated clinical and basic research approaches in cell therapy, immunology, and organ transplant genetics, applying statistical modeling to large-scale patient datasets.

Biorepository Technician

Sep 2014 – Jan 2016

National Marrow Donor Program, Research Repository

New Brighton, MN

- Managed 300k patient database and long-term storage for FDA and Clinical Trials Network samples using Good Laboratory Practice (GLP) standards, ensuring data integrity and regulatory compliance.
- Conducted high-throughput specimen processing in clinical laboratory facility, processing 1000s of samples per day while maintaining rigorous quality control and data tracking protocols.

TECHNICAL ACHIEVEMENTS

Established FAIR-compliant data governance framework for 0.5PB+ genomic datasets across international consortium
 Developed automated ID harmonization platform (IDhub) unifying 20 years of longitudinal study data
 Implemented standardized statistical methodology (CoreSC) adopted across 15+ research projects
 Reduced genomic annotation processing time from days to hours through optimized analytics platform deployment
 Led data team of 4 scientists coordinating analysis efforts across 650+ global collaborators
 Achieved zero data loss incidents over 4+ years while managing enterprise-scale research datasets
 Contributed to \$9.5M+ in successful grant awards through data management expertise and strategic analytics
 Delivered \$250K+ annual cost savings through strategic analytics platform optimization
 Pioneered enterprise AI integration deployment and secure LLM access for bioinformatics team workflows
 Developed official institutional nf-core nextflow execution profile accepted as open-source contribution

DATA SCIENCE & ANALYTICS EXPERTISE

Data Science & Machine Learning: Statistical modeling, Machine learning, Predictive analytics, PCA, UMAP, t-SNE, Hidden Markov Models, Automated cell typing, Multi-omic integration, Pattern recognition
Programming & Analysis: Python (pandas, numpy, scipy, matplotlib, seaborn, scikit-learn), R (ggplot2, dplyr, Seurat, rmarkdown, shiny), SQL, BASH/Shell, Statistical computing
Data Governance & Compliance: FAIR data principles, HIPAA compliance, NIH data sharing, PHI data management, Data stewardship, Regulatory frameworks, Audit logging, Data version control (DVC)
Cloud Analytics Platforms: AWS (EC2, S3, RDS, CloudFront), PostgreSQL, MySQL, Oracle, REDCap, LabKey, Open Science Framework, Dataverse, ETL frameworks
Genomics Analytics: Single-cell RNA-seq, Spatial transcriptomics, WGS/WES analysis, GWAS, Copy number variation, Multi-omic data integration, Genomic annotation, Variant analysis
Platform Development: REST API development, Database design, Automated workflows, Containerization (Docker/Kubernetes), CI/CD pipelines, Metadata management systems
Visualization & Reporting: Interactive dashboards, Publication-quality figures, Web applications, Data storytelling, Executive reporting, Scientific communication

EDUCATION

University of Minnesota

Minneapolis, MN

Master of Science in Bioinformatics and Computational Biology

2021

University of Minnesota

Minneapolis, MN

Bachelor of Science in Biochemistry, College of Biological Sciences

2012